Module 5 – Linear Transformations

5.1

Linear Transformation

Definition: Let $U(F)$ and $V(F)$ be two vector spaces. Then the function $T: U \longrightarrow V$ is called a linear transformation from U into V if

$$T(a\alpha + b\beta) = aT(\alpha) + bT(\beta) \text{ for all } a, b \in F \text{ and } \alpha, \beta \in U.$$

Note: ① If T is a linear transformation from U into itself (i.e., $T: U \to U$), then T is called a Linear operator on U.

② If $T: U \longrightarrow F$ is a linear transformation, then T is called a linear functional on U.

③ An n dimensional vector space V over a field F is denoted by $V_n(F)$.

Properties of Linear Transformation.

Let $T: U(F) \longrightarrow V(F)$ be a linear transformation from the vector space $U(F)$ to the vector space $V(F)$. Then

(i) $T(\hat{0}) = \bar{0}$ where $\hat{0} \in U$ and $\bar{0} \in V$

(ii) $T(-\alpha) = -T(\alpha)$ for all $\alpha \in U$

(iii) $T(\alpha - \beta) = T(\alpha) - T(\beta)$ for all $\alpha, \beta \in U$

(iv) $T(a_1\alpha_1 + a_2\alpha_2 + \cdots + a_n\alpha_n) = a_1T(\alpha_1) + a_2T(\alpha_2) + \cdots + a_nT(\alpha_n)$ for $a_1, a_2, \ldots, a_n \in F$ and $\alpha_1, \alpha_2, \ldots, \alpha_n \in U$.

Problems

① The mapping $T: V_3(\mathbb{R}) \longrightarrow V_2(\mathbb{R})$ is defined by $T(x,y,z) = (x-y,\ x-z)$ for $(x,y,z) \in V_3$. Is T a linear transformation?

Sol: Let $\alpha = (x_1, y_1, z_1)$ and $\beta = (x_2, y_2, z_2)$ be any two vectors in V_3.

For $a, b \in \mathbb{R}$,

$$T(a\alpha + b\beta) = T(a(x_1,y_1,z_1) + b(x_2,y_2,z_2))$$

$$= T(ax_1+bx_2,\ ay_1+by_2,\ az_1+bz_2)$$

$$= \left((ax_1+bx_2) - (ay_1+by_2),\ (ax_1+bx_2) - (az_1+bz_2)\right)$$

$$= \left(a(x_1-y_1) + b(x_2-y_2),\ a(x_1-z_1) + b(x_2-z_2)\right)$$

$$= a(x_1-y_1,\ x_1-z_1) + b(x_2-y_2,\ x_2-z_2)$$

$$= aT(x_1,y_1,z_1) + bT(x_2,y_2,z_2)$$

$$= aT(\alpha) + bT(\beta).$$

Therefore, T is a linear transformation.

2. Is the mapping $T: \mathbb{R}^3(\mathbb{R}) \longrightarrow \mathbb{R}^2(\mathbb{R})$ defined by $T(x,y,z) = (|x|, 0)$ a linear transformation?

Sol: Let $\alpha = (x_1, y_1, z_1)$, $\beta = (x_2, y_2, z_2) \in \mathbb{R}^3$.

For $a, b \in \mathbb{R}$, $T(a\alpha + b\beta) = T(a(x_1,y_1,z_1) + b(x_2,y_2,z_2))$

$= T(ax_1+bx_2,\ ay_1+by_2,\ az_1+bz_2)$

$= (|ax_1+bx_2|, 0)$

and $aT(\alpha)+bT(\beta) = aT(x_1, y_1, z_1) + bT(x_2, y_2, z_2)$

$= a(|x_1|, 0) + b(|x_2|, 0)$

$= (a|x_1| + b|x_2|, 0)$

Clearly $T(a\alpha+b\beta) \neq aT(\alpha)+bT(\beta)$

(since $|ax_1+bx_2| \leq |a||x_1|+|b||x_2|$)

Hence T is not a linear transformation.

3. Is $T: \mathbb{R}^3(\mathbb{R}) \longrightarrow \mathbb{R}^3(\mathbb{R})$ defined by

$T(x, y, z) = (x+1, y, z)$ a linear transformation

<u>Answer:</u> not a L.T.

<u>Note:</u> Let $U(F)$ and $V(F)$ be two vector spaces and $S = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ a basis of U. Let $\{\beta_1, \beta_2, \ldots, \beta_n\}$ be a set of vectors in V. Then there exists a unique linear transformation $T: U \longrightarrow V$ such that $T(\alpha_i) = \beta_i$ for $i = 1, 2, \ldots, n$

4. Fin a linear Transformation $T : \mathbb{R}^2(\mathbb{R}) \to \mathbb{R}^2(\mathbb{R})$ is defined by $T(2,3) = (4,5)$ and $T(1,0) = (0,0)$.

Sol: Let $S = \{(2,3), (1,0)\}$. First we have to show that S is a basis of $\mathbb{R}^2$.

Let $(x, y) \in \mathbb{R}^2$. For $a, b \in \mathbb{R}$,

$$(x, y) = a(2,3) + b(1,0)$$
$$= (2a+b, 3a)$$
$$\Rightarrow 2a+b = x \text{ and } 3a = y$$
$$\Rightarrow a = \frac{y}{3} \text{ and } b = \frac{3x-2y}{3}$$
$$\therefore (x,y) = \frac{y}{3}(2,3) + \left(\frac{3x-2y}{3}\right)(1,0)$$

and hence $L(S) = \mathbb{R}^2$. Clealy S is L.I.

Therefore, S is a basis of $\mathbb{R}^2$.

Now, $$T(x,y) = T\left(\frac{y}{3}(2,3) + \left(\frac{3x-2y}{3}\right)(1,0)\right)$$

$$= \frac{y}{3} T(2,3) + \frac{3x-2y}{3} T(1,0)$$

$$= \frac{y}{3}(4,5) + \frac{3x-2y}{3}(0,0)$$

$$= \left(\frac{4y}{3}, \frac{5y}{3}\right)$$

which is the required linear transformation

5. Find a linear transformation $T: \mathbb{R}^3(\mathbb{R}) \longrightarrow \mathbb{R}(\mathbb{R})$ defined by $T(1,1,1) = 3$, $T(0,1,-2) = 1$, $T(0,0,1) = -2$.

Answer: $T(x,y,z) = 8x - 3y - 2z$.

6. Find a linear transformation $T: V_2(\mathbb{R}) \longrightarrow V_2(\mathbb{R})$ defined by $T(1,2) = (3,0)$ and $T(2,1) = (1,2)$.

Answer: $T(x,y) = \left(\frac{5y-x}{3}, \frac{4x-2y}{3} \right)$.

Range space and Null space of a Linear Transformation

Definition: Let $U(F)$ and $V(F)$ be vectorspaces and $T: U \longrightarrow V$ a linear transformation. Then the range of T is denoted by $R(T)$ and defined as

$$R(T) = \{ T(\alpha) \mid \alpha \in U \}$$

clearly $R(T)$ is a subspace of $V(F)$.

Definition: Let $U(F)$ and $V(F)$ be vector spaces and $T: U \longrightarrow V$ a linear transformation. Then the Null space of T is denoted by $N(T)$ and defined as $N(T) = \{ \alpha \in U \mid T(\alpha) = \hat{0} \}$, $\hat{0}$ is the zero vector in V.

Clearly $N(T)$ is a subspace of $U(F)$.

The Null space $N(T)$ is also called as the Kernel of T.

Dimension of Range space and Null space

Definition: Let $T: U(F) \longrightarrow V(F)$ be a linear transformation where U is a finite dimensional vector space. Then

(i) The dimension of the range space $R(T)$ is defined as rank of T and this is denoted by $\rho(T)$. i.e., $\rho(T) = \dim R(T)$.

(ii) The dimension of the Null space $N(T)$ is defined as nullity of T and is denoted by $\nu(T)$. i.e., $\nu(T) = \dim N(T)$.

Note: $\boxed{\text{rank}(T) + \text{nullity}(T) = \dim U}$.

Problems:

① Let $T: \mathbb{R}^3(\mathbb{R}) \longrightarrow \mathbb{R}^2(\mathbb{R})$ be a linear transformation defined by $T(x, y, z) = (x+y,\ y+z)$. Then find the dimension of range space $R(T)$ and the null space $N(T)$.

Sol: Let $S = \{(1,0,0), (0,1,0), (0,0,1)\}$ be the standard basis of $\mathbb{R}^3$.

Then $T(1,0,0)=(1,0)$

$T(0,1,0)=(1,1)$

$T(0,0,1)=(0,1)$

Let $S_1=\{(1,0),(1,1),(0,1)\}$. Then $S_1 \subseteq R(T)$.

Now, $\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \xrightarrow{R_2-R_1} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \xrightarrow{R_3-R_1} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$

Therefore, $\{(1,0),(0,1)\}$ forms a basis of $R(T)$.

Hence $\dim R(T) = 2$.

For $\alpha \in N(T)$, $T(\alpha)=\hat{0} \Rightarrow T(x,y,z)=(0,0)$

$\Rightarrow (x+y,\ y+z)=(0,0)$

$\Rightarrow x+y=0$

and $y+z=0$

Let $z=k$, then we get $y=-k$ and hence

$x=k,\ y=-k,\ z=k$.

$\therefore \left\{\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}\right\}$ form a basis for $N(T)$ and hence

$\dim N(T) = \text{nullity}(T) = 1$

Therefore, $\dim R(T) + \dim N(T) = 2+1$

i.e., $\boxed{\text{rank}(T) + \text{nullity} = 3 = \dim \mathbb{R}^3}$

2. Let $T: \mathbb{R}^4(\mathbb{R}) \longrightarrow \mathbb{R}^3(\mathbb{R})$ be a linear transformation defined by $T(a,b,c,d)=(a-b+c+d,\ a+2c-d,\ a+b+3c-3d)$ for $a,b,c,d \in \mathbb{R}$, then find basis and dimension of $R(T)$ and $N(T)$. Answer: (i) $\{(1,1,1),(0,1,2)\}$ is a basis of $R(T)$

$\dim R(T)=2$, $\dim N(T)=\text{nullity}(T)=2$ (ii) $\{(-2,-1,1,0),(1,2,0,1)\}$ is a basis of ...

Invertible Linear Transformation

Non-Singular Transformation :

Definition : A linear transformation $T : U(F) \to V(F)$ is said to be non-singular if the null space of T consists zero vector alone. [i.e., $N(T) = \{\hat{0}\}$, where $\hat{0}$ is the zero vector in $U(F)$.]

i.e, $\alpha \in U$ and $T(\alpha) = \bar{0} \Rightarrow \alpha = \hat{0}$.

Note: ① Let $U(F)$ and $V(F)$ be two finite dimensional vector spaces. The linear transformation $T : U \to V$ is one-one onto iff (if and only if) T is non-singular. So, $T : U \to V$ is invertible if and only if T is non-singular.

② A linear transformation $T : U(F) \to V(F)$ is said to be invertible if there exists a linear transformation $S : V(F) \to U(F)$ such that

$$T \circ S = S \circ T = I, \text{ identity transformation}$$

Here S is the inverse of T.

③ Let $U(F)$ and $V(F)$ be two vector spaces. Then the set $L(U, V)$ of all linear transformations from $U(F)$ into $V(F)$ is a vector space over a field F. This is called a vector space of linear transformations.

④ If $\dim U(F) = m$ and $\dim V(F) = n$, then $\dim(L(U,V)) = mn$.

Problems

① If a linear transformation $T: \mathbb{R}^3(\mathbb{R}) \longrightarrow \mathbb{R}^3(\mathbb{R})$ defined by $T(x,y,z) = (2x, (4x-y), (2x+3y-z))$ is invertible, then find T^{-1}.

Sol: Let $T(x,y,z) = (a,b,c)$

Then $T^{-1}(a,b,c) = (x,y,z)$

(since T is invertible)

Now, $T(x,y,z) = (a,b,c)$

$\Rightarrow (2x, 4x-y, 2x+3y-z) = (a,b,c)$

$\Rightarrow 2x = a, \; 4x - y = b, \; 2x + 3y - z = c$

$\Rightarrow x = \frac{a}{2}, \; y = 2a - b, \; z = 7a - 3b - c$

$\therefore T^{-1}(a,b,c) = \left(\frac{a}{2}, 2a-b, 7a-3b-c\right)$

② Show that a linear transformation $T: \mathbb{R}^3(\mathbb{R}) \longrightarrow \mathbb{R}^3(\mathbb{R})$ defined by $T(x,y,z) = (x+y+z, y+z, z)$ is invertible and hence find T^{-1}.

Sol: We have T is invertible if T is non-singular

For $\alpha = (x, y, z) \in \mathbb{R}^3$, $T(\alpha) = \hat{0}$

$$\Rightarrow T(x, y, z) = (0, 0, 0)$$
$$\Rightarrow (x+y+z,\ y+z,\ z) = (0, 0, 0)$$
$$\Rightarrow x+y+z = 0,\ y+z = 0,\ z = 0$$
$$\Rightarrow x = 0,\ y = 0,\ z = 0$$
$$\Rightarrow \alpha = \hat{0}$$

Therefore, $N(T) = \{\hat{0}\}$ and hence T is non-singular. Thus T is invertible.

Let $T(x, y, z) = (a, b, c)$. Then $T^{-1}(a, b, c) = (x, y, z)$

Now, $T(x, y, z) = (a, b, c) \Rightarrow (x+y+z,\ y+z,\ z) = (a, b, c)$

$$\Rightarrow x+y+z = a,\ y+z = b,\ z = c$$
$$\Rightarrow x = a-b,\ y = b-c,\ z = c.$$
$$\therefore T^{-1}(a, b, c) = (a-b,\ b-c,\ c)$$

3. Show that a linear transformation $T: \mathbb{R}^3(\mathbb{R}) \to \mathbb{R}^3(\mathbb{R})$ defined by $T(x, y, z) = (3x,\ x-y,\ 2x+y+z)$ is invertible and hence find T^{-1}.

4. Show that a linear transformation $T: \mathbb{R}^2(\mathbb{R}) \to \mathbb{R}^2(\mathbb{R})$ defined by $T(x, y) = (x+y,\ x-y)$ is invertible and ~~(and)~~ hence find T^{-1}.

5.2 Matrix of Linear Transformation

Let $U(F)$ and $V(F)$ be two finite dimensional vector spaces such that $\dim U = n$ and $\dim V = m$. Let $T: U \to V$ be a linear transformation. Let $B_1 = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ be the ordered basis of U and $B_2 = \{\beta_1, \beta_2, \dots, \beta_m\}$ be the ordered basis of V.

For every $\alpha \in U$, $T(\alpha) \in V$ and $T(\alpha)$ can be expressed as a linear combination of the elements of the basis B_2. So,

$$T(\alpha_1) = a_{11}\beta_1 + a_{21}\beta_2 + \dots + a_{m1}\beta_m$$

$$T(\alpha_2) = a_{12}\beta_1 + a_{22}\beta_2 + \dots + a_{m2}\beta_m$$

$$\vdots$$

$$T(\alpha_n) = a_{1n}\beta_1 + a_{2n}\beta_2 + \dots + a_{mn}\beta_m$$

writing the co-ordinates of $T(\alpha_1), T(\alpha_2), \dots, T(\alpha_n)$ successively as columns of a matrix, we get

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}.$$

This matrix represented as $[a_{ij}]_{m \times n}$ is called the matrix of the linear transformation T with respect to the bases B_1 and B_2. we can denote this by $[T : B_1, B_2]$ or $[T] = [a_{ij}]_{m\times n}$.

Here $\begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}, \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix}, \dots, \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}$ are

co-ordinate vectors of $T(\alpha_1), T(\alpha_2), \dots, T(\alpha_n)$

with respect to a basis B_2 respectively.

i.e., $[T(\alpha_1)]_{B_2} = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}, [T(\alpha_2)]_{B_2} = \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix}, \dots, [T(\alpha_n)]_{B_2} = \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}$

Note: (i) Let $T : V(F) \longrightarrow V(F)$ be a linear operator such that $\dim V = n$. If $B_1 = B_2 = B$ (say), then the above said matrix is called the matrix of T relative to the ordered basis B of V. It is denoted by $[T : B]$ or $[T]_B$.

(ii) Let $B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ be a basis of an n-dimensional vector space $V(F)$. Then for any $\alpha \in V$, $\alpha = a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n$, where $a_1, a_2, \dots, a_n \in F$.

Therefore, the co-ordinate vector of α relative to the basis B is $[\alpha]_B = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$.

Problems

1. Let $T: \mathbb{R}^2(\mathbb{R}) \longrightarrow \mathbb{R}^3(\mathbb{R})$ be a linear transformation defined by $T(x,y) = (x+y,\ 2x-y,\ 7y)$. Find the matrix of linear transformation $[T : B_1, B_2]$ where B_1 is the standard basis of $\mathbb{R}^2$ and B_2 is the standard basis of $\mathbb{R}^3$.

Sol: We have $B_1 = \{(1,0),(0,1)\}$ is the standard basis of $\mathbb{R}^2$ and $B_2 = \{(1,0,0),(0,1,0),(0,0,1)\}$ is the standard basis of $\mathbb{R}^3$.

Now,
$$T(1,0) = (1,2,0)$$
$$= 1(1,0,0) + 2(0,1,0) + 0(0,0,1)$$
$$T(0,1) = (1,-1,7)$$
$$= 1(1,0,0) - 1(0,1,0) + 7(0,0,1).$$

Therefore,
$$[T : B_1, B_2] = \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ 0 & 7 \end{bmatrix}.$$

2. Let $T: \mathbb{R}^3(\mathbb{R}) \longrightarrow \mathbb{R}^2(\mathbb{R})$ be a linear transformation defined by $T(x,y,z) = (3x+2y-4z,\ x-5y+3z)$. Find the matrix of T relative to the bases

$$B_1 = \{(1,1,1),(1,1,0),(1,0,0)\} \text{ and } B_2 = \{(1,3),(1,5)\}.$$

Sol: Let $(a,b) \in \mathbb{R}^2$ be such that
$$(a,b) = p(1,3) + q(2,5)$$
then $(a,b) = (p+2q,\ 3p+5q)$
$$\Rightarrow a = p+2q,\ b = 3p+5q$$
$$\Rightarrow p = -5a+2b,\ q = 3a-b$$

$\therefore\ (a,b) = (-5a+2b)(1,3) + (3a-b)(2,5)$.

Now, $T(1,1,1) = (1,-1)$

$= -7(1,3) + 4(2,5)$

$T(1,1,0) = (5,-4)$

$= -33(1,3) + 19(2,5)$

$T(1,0,0) = (3,1)$

$= -13(1,3) + 8(2,5)$.

Therefore, the matrix of T relative to B_1 and B_2 is $[T : B_1, B_2] = \begin{bmatrix} -7 & -33 & -13 \\ 4 & 19 & 8 \end{bmatrix}$

* Suppose, $\alpha = (1,1,1)$, then $[T(\alpha)]_{B_2} = \begin{bmatrix} -7 \\ 4 \end{bmatrix}$ is the co-ordinate vector of $T(\alpha)$ relative to B_2

3. Let $T : \mathbb{R}^2(\mathbb{R}) \longrightarrow \mathbb{R}^2(\mathbb{R})$ be a linear operator defined by $T(x,y) = (4x-2y, 2x+y)$. Find the matrix of T w.r.t. the basis $B = \{(1,1), (-1,0)\}$.

Sol: let $(a,b) \in \mathbb{R}^2$ be such that

$$(a,b) = p(1,1) + q(-1,0)$$

then $(a,b) = \cancel{(p-q) + p}\ (p-q, p)$

$$\Rightarrow a = p-q, \quad b = p$$

$$\Rightarrow p = b, \quad q = b-a$$

$$\therefore (a,b) = b(1,1) + (b-a)(-1,0)$$

Now, $T(1,1) = (2,3)$

$$= 3(1,1) + 1(1,0)$$

$$T(-1,0) = (-4,-2)$$

$$= -2(1,1) + 2(-1,0)$$

Therefore, the matrix of T w.r.t. the basis

$B = \{(1,1), (-1,0)\}$ is

$$[T:B] = [T]_B = \begin{bmatrix} 3 & -2 \\ 1 & 2 \end{bmatrix}$$

* Here $\begin{bmatrix} 3 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 2 \end{bmatrix}$ are the co-ordinate

vectors

5.3 Change of Basis

Let $V(F)$ be an n-dimensional vector space.
Let $B = \{\alpha_1, \alpha_2, \cdots, \alpha_n\}$ and $B' = \{\beta_1, \beta_2, \cdots, \beta_n\}$ be two arbitrary bases for V. Let us suppose

$$\beta_1 = a_{11}\alpha_1 + a_{21}\alpha_2 + \cdots + a_{n1}\alpha_n$$
$$\beta_2 = a_{12}\alpha_1 + a_{22}\alpha_2 + \cdots + a_{n2}\alpha_n$$
$$\vdots$$
$$\beta_n = a_{1n}\alpha_1 + a_{2n}\alpha_2 + \cdots + a_{nn}\alpha_n$$

Then the matrix of transformation from B to B' is

$$P = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & & a_{nn} \end{bmatrix}.$$

Here, the matrix P is called the transition matrix from B to B'.
(or, the change of basis matrix)

Note: Since $B' = \{\beta_1, \beta_2, \ldots, \beta_n\}$ is L.I, the matrix P is invertible and P^{-1} is called the transition matrix from B' to B.

Examples

1. Let $B = \{(1,0),(0,1)\}$ and $B' = \{(1,3),(2,5)\}$ be the bases of $\mathbb{R}^2(\mathbb{R})$. Find the transition matrix from B to B'. Also find the transition matrix from B' to B.

Sol: (i) clearly, $(1,3) = 1(1,0) + 3(0,1)$

$$(2,5) = 2(1,0) + 5(0,1)$$

Therefore, the transition matrix from B to B' is $\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$.

(ii) For any $(x,y) \in \mathbb{R}^2$,

$$(x,y) = a(1,3) + b(2,5)$$

$$\Rightarrow x = a + 2b, \quad y = 3a + 5b$$

$$\Rightarrow a = 2y - 5x; \; b = 3x - y.$$

$$\therefore (x,y) = (2y - 5x)(1,3) + (3x - y)(2,5)$$

Now,

$$(1,0) = -5(1,3) + 3(2,5)$$

$$(0,1) = (+2)(1,3) - 1(2,5)$$

Therefore, the transition matrix from B' to B is $\begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix}$.

(2) Find the transition matrix from $B=\{(1,2),(3,4)\}$ to $B'=\{(7,3),(4,2)\}$, where B and B' are bases of a vector space $\mathbb{R}^2(\mathbb{R})$.

(3) Find the transition matrix from $B=\{(9,2),(4,-3)\}$ to $B'=\{(2,1),(-3,1)\}$, where B and B' are the bases of $\mathbb{R}^2(\mathbb{R})$.

(4) Find the transition matrix from $B=\{(1,0,0),(0,1,0),(0,0,1)\}$ to $B'=\{(1,1,0),(0,1,2),(1,1,1)\}$, where B and B' are the bases of $\mathbb{R}^3(\mathbb{R})$.

Similarity of Matrices

__Definition__: Let A and B be square matrices of order n over a field F. Then B is said to be similar to A if there exists an invertible matrix C of order n with elements in F such that $B=C^{-1}AC$.